MAJOR PROJECT REPORT

B.Tech in Computer Science and Engineering

National Institute of Technology Karnataka, Surathkal

TrustLens

A Semi-Automated Framework for Trustworthiness Assessment and Auditing of Hugging Face Text-Classification Models

A framework organised around the five FRIES dimensions for auditing Hugging Face text-classification models:
Fairness, Robustness, Integrity, Explainability and Safety

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Repository: `https://github.com/Aayushman00/TrustLens`

Date of submission:

Certificate

This is to certify that the major project report entitled **"TrustLens: A Semi-Automated Framework for Trustworthiness Assessment and Auditing of Hugging Face Text-Classification Models"** is a record of work carried out by **Aayushman (231CS105), Ashutosh Kumar (231CS113)** and **Chetan Kumar Sah (231CS118)** under my supervision, in partial fulfilment of the requirements for the award of the degree of Bachelor of Technology in Computer Science and Engineering at National Institute of Technology Karnataka, Surathkal, during the academic year 2026–27.

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Place: Surathkal

Date: 25 October 2026

Declaration

We declare that this report describes our own work on the TrustLens project. The software, the experimental models, the review packets and the evaluation runs reported here were produced by the three of us, and the numbers in the results chapters are taken from result files stored in the project repository. The human-review data in Chapter 6 come from volunteer reviewers who filled in an anonymous online form; we did not write or edit any of their answers. Where we rely on other people's ideas, datasets or models, we cite them in the reference list.

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Acknowledgement

We thank our supervisor, Mr. Alwyn Roshan Pais, Professor, Department of CSE, for steering us away from a purely engineering project and towards questions that we could actually test. The most useful push was a simple one: does the tool behave differently on models we know are worse, and if it does, does it still do so when somebody else looks at the same evidence? We are also grateful to the maintainers of the Hugging Face Hub, of the `transformers` and `datasets` libraries and of the Civil Comments dataset, whose open releases made it possible to run everything on one laptop GPU. Finally, thanks to the people who gave their time to review the evidence packets, and to those who read early drafts of the experiment design and found holes in it.

Abstract

Downloading a classifier from a public model hub takes one line of code, but deciding whether it deserves to be used is harder: the information at hand is a free-text model card, a possibly empty licence field and no standard evidence about behaviour on sensitive groups. **TrustLens** is a semi-automated research prototype that collects this evidence for Hugging Face text-classification models and organises it around the five FRIES dimensions: Fairness, Robustness, Integrity, Explainability and Safety. It imports a model, runs five automated tests (probes) locally, stores the evidence in files named by their SHA-256 hashes, converts it into Occurrence/Severity/Detection (O/S/D) ratings, three scales borrowed from Failure Mode and Effects Analysis (FMEA), and produces a versioned report. The primary output is a five-dimensional audit profile; the aggregate FRIES score is a secondary summary. The behavioural experiments cover one task, English-language toxicity classification on the Civil Comments dataset.

We evaluated the system with controlled flawed models, experiments with progressively larger injected defects, a fairness confirmation planned before it was run, a comparison with interpretation by a large language model (LLM), a human evidence review and a descriptive audit of seven Hub models. Measured evidence from the deterministic probes was identical across five repeated runs and changed consistently with the injected defect level in every experiment that returned a measurement. A scoring method scaled to the range of the injected defects widened the score range from 0.68–1.18 to 4.60–6.65 points. Separating missing documentation from evidence of an observed problem removed false flags on clean controls, and a rule on the gap in true-positive rates between two groups (the equal-opportunity difference) flagged every injected disparity at label-flip rates of 0.4 and above on fresh data. Several results were negative: the prespecified robustness sweep withheld a rating at every level, the fairness rule also flagged the real reference model, and the comparison with LLM interpretation was inconclusive (22 of 60 planned answers). The scores separate synthetic defect levels but are not calibrated to real-world harm; the fairness groups come from an annotator-reported identity-attack signal, with no mention-based rerun; and the human review and Hub audit are exploratory and descriptive (Section 7.3).

TrustLens is meant to help a human reviewer see where the evidence is strong and where it is thin.

Keywords: model auditing, trustworthy AI, FRIES, Hugging Face Hub, fairness, equal opportunity, artifact integrity, model cards, text classification.

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1 Introduction

1.1 Problem and gap

Most people who need a toxicity classifier do not train one. They search the Hugging Face Hub, skim a model card, check the download count and call `from_pretrained`. That is the point where trust is decided almost by accident. The Hub holds a very large number of checkpoints uploaded by anyone with an account, and an empirical study found that their metadata and documentation are often incomplete or inconsistent [16]. A card can be an untouched template, a licence field can be empty or contradict the card, and a card can claim "no known biases" while the model's behaviour on identity-related text says otherwise.

Auditing such a model is difficult for three reasons. Trust has several dimensions that accuracy does not capture [5, 6]. The only standard disclosure, the model card, is voluntary and unverified. And existing scoring ideas leave the hard step open: the FRIES trust score [1] combines Occurrence, Severity and Detection (O/S/D) ratings into one 0–10 number, but leaves the step from *measured evidence* to those ratings to experts. A tool has to define that step and show how weak the definition is.

This project asks a narrow question: *given a model reference and a small labelled dataset, can a tool collect evidence about the model reproducibly, and show a reviewer where the risks are and where the evidence is too thin to say?* We do not ask whether a model is "trustworthy" in an absolute sense; we describe what the tests can see.

1.2 Research questions

RQ1 Does the pipeline collect the evidence it claims to collect, reproducibly?

RQ2 When a defect of known severity is injected into a model, do the measured evidence, the risk decision and the score each respond, and in proportion to severity?

RQ3 Where the decision layers fail to respond, can the failure be fixed without tuning against the data that exposed it?

RQ4 How does deterministic interpretation of evidence compare with LLM-based interpretation of the same evidence?

RQ5 Do independent evidence reviewers agree with each other, and with TrustLens?

RQ6 What does an audit profile look like for real Hub models, and what happens when the evidence is insufficient?

1.3 Contributions

TrustLens integrates known techniques (group-fairness metrics, a perturbation test, a miss-rate test, model-card checks, file hashing) under the FRIES formula. We do not claim novel algorithms. The contribution is an applied framework and an empirical investigation of how its evidence, decision and scoring layers behave:

1. **An integrated audit workflow** for Hugging Face text-classification models, organised around Fairness, Robustness, Integrity, Explainability and Safety, from model import

to a versioned report.
2. **Separation of measured evidence, risk decisions and ratings**, with explicit withholding of a rating when evidence is insufficient.
3. **Missing documentation treated separately** from evidence of an observed problem ("disclosure gaps": information a model card should give but does not).
4. **Controlled experiments** on repeatability, defect detection, score sensitivity and decision behaviour, including negative results.
5. **File-set verification and evidence provenance**: every file of a model is compared with a reference hash, every probe output is stored under its SHA-256 hash, and every evaluation records the methodology version and exact model version.
6. **An evaluation** with controlled flawed models, blinded evidence reviewers and seven real Hub models.

1.4 Scope and intended use

TrustLens is a research prototype for local, single-user use. Its behavioural experiments cover one task: English-language toxicity classification on Civil Comments. The five-dimensional audit profile is the primary output; the aggregate FRIES score is a secondary summary. TrustLens supports a human reviewer in seeing where evidence is strong and where it is thin. It is not a certification, ranking or deployment-approval tool, and a measured property is evidence about trust, not trust itself. Chapter 2 reviews related work, Chapters 3–4 describe the system and method, Chapters 5–6 the experiments and results, and Chapters 7–9 the interpretation, limits and conclusions.

2 Literature Review

2.1 Trustworthiness and the FRIES score

Early work treated trust as accuracy and generalisation; later surveys [5, 6] and policy documents such as the NIST AI Risk Management Framework [7] list fairness, robustness, transparency, safety, accountability and human oversight, and stress that no single metric captures trust and that trade-offs need human judgment. Vashistha and Farahi tie trust to calibration [3] and to expected utility [4]; their work concerns probabilistic classifiers and does not cover documentation, provenance or file integrity, where much of the practical risk on a public hub sits.

Rutinowski et al. [1] propose FRIES as an FMEA-inspired trust score. For each risk an analyst rates Occurrence, Severity and Detection; per-risk scores are aggregated per dimension (Fairness, Robustness, Integrity, Explainability, Safety) and across dimensions. The paper leaves open how a concrete measurement, say a parity gap of 0.17 with a wide interval, becomes an integer rating. That gap is the main problem on the implementation side of this project.

2.2 Measures behind each dimension

Fairness. Group-fairness criteria compare behaviour across groups: demographic parity compares positive-prediction rates, equal opportunity and equalized odds [10] compare true- and false-positive rates. Borkan et al. [11] showed that toxicity classifiers pick up identity terms as cues and released the Civil Comments data with identity annotations used here; Mehrabi et al. [12] survey how bias enters a pipeline. **Robustness.** HotFlip [13] and CheckList [14] are far stronger than the TrustLens probe, which swaps a few random characters and measures typo sensitivity, not adversarial robustness. **Explainability and transparency.** We follow the documentation-centred line of model cards [8] and datasheets [9]: the probe examines what a card discloses, not model internals. **Supply chain.** Reuse of pre-trained models creates a software supply chain with documentation and attribution risks [16], and a tampered checkpoint can hide a backdoor [15].

2.3 Existing systems and positioning

TrustML [2] computes composite trust indicators from a catalogue of user-chosen metrics, with no model import and no documentation or integrity evidence. The FRIES paper is a methodology, not a tool, and the I/U-trustworthy frameworks cover probabilistic classifiers only. Model cards [8] are voluntary and unverified. TrustLens is an *applied integration* of known pieces (group-fairness metrics, a perturbation test, a miss-rate test, card checks, file hashing) under the FRIES formula: it imports a Hub model at an exact (pinned) revision, runs local probes, stores hashed evidence, proposes O/S/D ratings and supports review. Its probes are narrow (one task) and its O/S/D mapping is an unvalidated proposal. We found no public tool that does exactly this for Hub models, but our search was not systematic, so we claim no novelty beyond that.

3 System Architecture and Implementation

3.1 Conceptual flow

TrustLens turns a model and a labelled dataset into an audit report in three stages that are kept apart on purpose (Figure 3.1). Five probes produce *measured evidence*. A *risk assessment* decides, from that evidence, whether a risk is present, whether only a documentation gap exists, or whether the evidence is too thin to decide. Only then are O/S/D *ratings* proposed or entered by a reviewer and combined into dimension and aggregate scores. A probe or rating that lacks sufficient evidence is *withheld* (the audit records that the evidence was insufficient and gives no rating) rather than guessed, and the aggregate score is withheld whenever any dimension lacks a complete O/S/D triple.

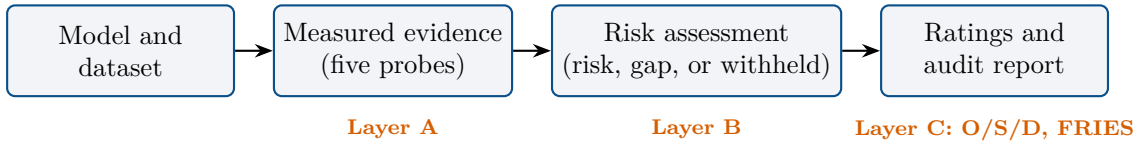


Figure 3.1: Conceptual flow of an audit. Layer B can withhold a rating; Layer C is computed only from ratings that exist.

3.2 Principal components

Figure 3.2 shows the deployment: a conventional web application with an asynchronous worker. Table 3.1 lists the principal components and their roles. Source-file mappings, enumerations and status codes are in `docs/ENGINEERING_NOTES.md`.

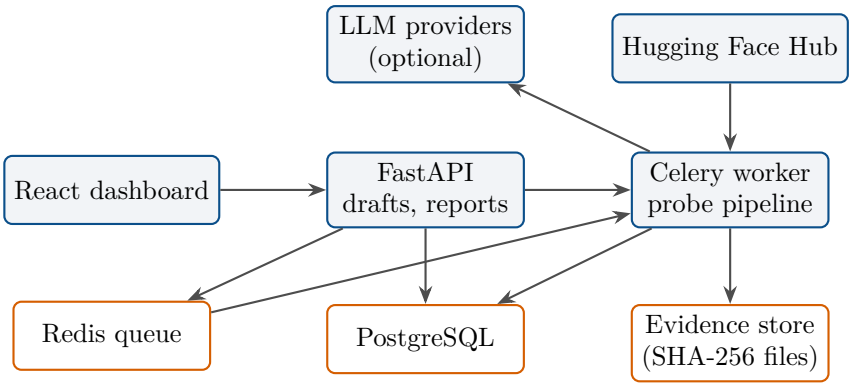


Figure 3.2: Deployment architecture of TrustLens.

Two limits define the prototype's scope: behavioural Safety needs a user-supplied labelled dataset (there is no built-in unsafe-prompt suite), and Explainability checks documentation rather than explaining individual predictions.

Table 3.1: Principal components.

Component	Role
Model import and local inference	Reads card, tags, licence and file list from the Hub without weights and fixes the revision; runs batched `transformers` inference [18] on GPU or CPU.
Five probes	Automated tests for Fairness, Robustness, Safety, Integrity and Explainability (Chapter 4); each returns metrics, a status and decision fields.
Assessment engines	*Deterministic* (default; proposes no ratings), *rule-based* (heuristic O/S/D proposals) and *LLM-assisted* (rule-based ratings plus an LLM re-rating of the card-based dimensions).
Scorer and reports	Computes dimension and aggregate FRIES scores from fixed test vectors; builds a versioned JSON and PDF report.
Human review	In human-review mode the pipeline pauses so a reviewer can edit the O/S/D ratings before scoring; in automatic mode it finalises without review and marks the report as not human-reviewed.
Evidence store	Raw probe output saved as JSON files named by their SHA-256 hash, in a local volume; PostgreSQL holds evaluations and scores.

3.3 Workflow, evidence handling and review

The API validates a request, stores an evaluation draft and enqueues a task; the worker loads the model, runs the probes, stores evidence, asks the assessment engine for ratings, scores them and writes the result. Each evaluation carries an *evaluation contract* that fixes the dataset (stored under its hash), the text, target and group columns, the mapping from dataset labels to model outputs, the label treated as favourable, the minimum group size and the exact model version. The contract made a label-mapping error visible in our first experiment (Section 5.1).

Reproducibility is built into the data model: evaluations are stamped with a methodology version, the loaded model snapshot is checked against the pinned revision (the exact model version recorded for the audit), probes take explicit seeds, and datasets are stored by content hash. A probe output that violates its contract fails the evaluation instead of being stored. LLM failures never fail an evaluation: rule-based fallback ratings are kept and the report records why.

The dashboard presents the five dimensions with their evidence and O/S/D source, and states the mode, the engine and whether a human reviewed the numbers, with the profile first and the aggregate last. Engineering detail, installation, the technology stack and dashboard screenshots are in `docs/ENGINEERING_NOTES.md` and the repository README.

4 FRIES Methodology

This chapter defines what TrustLens measures and how evidence becomes risk decisions, ratings and scores, as used in the experiments. Where the method changed during the project, only the variants needed to read a result are named; the full version history and the exact rule identifiers are in `docs/REPRODUCING_THESIS_RESULTS.md` and `docs/ENGINEERING_NOTES.md`.

4.1 From ratings to scores

FRIES [1] adapts FMEA. It covers five dimensions: *Fairness* (does behaviour differ between groups of texts), *Robustness* (does it survive small typos), *Integrity* (are the evaluated files and the model's identity what is claimed), *Explainability* (does the documentation explain the model) and *Safety* (does the model miss severe harmful content, and does its card disclose safety limits). For each relevant risk an analyst rates *Occurrence* (O: how unlikely the failure is), *Severity* (S: how mild its consequence is) and *Detection* (D: how easily it would be noticed). TrustLens uses the convention that **higher is better on all three scales**; the scorer requires integers in 0..10 and

$$\Pi_r = \sqrt[3]{O_r\, S_r\, D_r}, \qquad \Pi_r = 0 \text{ if any of } O_r, S_r, D_r = 0 \text{ (veto)}, \tag{4.1}$$

$$T_a = \frac{1}{|R_a|} \sum_{r \in R_a} \Pi_r, \qquad \text{FRIES} = \sum_a w_a\, T_a, \quad w_a = 0.2 \text{ by default}. \tag{4.2}$$

In every experiment each dimension has a single triple, so $T_a = \Pi_r$ and weights are equal. The geometric mean lets one low component dominate: $(1, 2, 2)$ gives $\Pi = 1.59$, $(9, 9, 8)$ gives 8.65. The rule-based engine proposes only 1..9 (0 and 10 are reserved for human judgment), so its dimension scores lie between 1.0 and 9.0.

4.2 Evidence: the five dimensions

4.2.1 Fairness

For a binary classifier with favourable class $\hat{y} = 1$ and groups g defined by a sensitive column, the probe computes the *demographic-parity difference* (DPD), the gap between groups in the share of texts predicted positive, and, for each group, the *true-positive rate* (TPR), the share of actual positives that are predicted positive. The gap in TPR between groups is the *equal-opportunity difference* (EOD):

$$\text{DPD} = \max_g P(\hat{Y}{=}1 \mid g) - \min_g P(\hat{Y}{=}1 \mid g), \tag{4.3}$$

$$\text{TPR}_g = \frac{TP_g}{TP_g + FN_g}, \qquad \text{EOD} = \max_g \text{TPR}_g - \min_g \text{TPR}_g. \tag{4.4}$$

The probe also reports the corresponding gap in false-positive rates (FPR, the share of actual negatives predicted positive) and the gap in label rates $P(Y{=}1 \mid g)$. EOD is the equal-opportunity difference of Hardt et al. [10]; it is the primary fairness quantity and is distinct from the larger equalized-odds difference (the larger of the TPR and FPR gaps).

Groups smaller than 30 are excluded. A TPR gap is evidence of a disparity that needs review; it is not by itself proof that a model is unfair.

Why the trigger is the TPR gap. A first rule based on "excess" demographic parity depended on group base rates $P(Y{=}1 \mid g)$: it was non-zero on clean controls and shrank as the injected defect grew. The TPR gap conditions on the true label. It was chosen after inspecting exploratory data, so it was confirmed on fresh data with a new seed (Section 5.3). The tolerance $\varepsilon = 0.02$, the largest TPR gap treated as negligible, is a method-wide practical tolerance and was not tuned.

Uncertainty is expressed as a 95% *confidence interval* (CI), a range around an estimated value: here a percentile bootstrap of EOD that re-samples the evaluation rows $B = 1000$ times (seed 42). The decision rule (Table 4.1) is a *disparity* rule, not a defect detector: a model with a real TPR gap but no injected defect is flagged, and the report calls this "evidence for review". The width rule often decides the outcome because most intervals in our data have width near 0.15; a rating withheld this way is not a pass.

Table 4.1: Decision rule of the binary fairness probe; the first matching row applies.

#	Condition	Finding	Risk
1	fewer than 2 eligible groups, a group without positives, or no CI	insufficient evidence	none
2	EOD CI width > 0.15	insufficient evidence (withheld)	none
3	EOD $> \varepsilon$ and CI lower bound $> \varepsilon$	fairness risk	fairness-gap risk (`F-FAIR-EOPP`)
4	EOD $> \varepsilon$, CI lower bound $\leq \varepsilon$	disparity observed	none
5	EOD $\leq \varepsilon$	no material disparity	none

4.2.2 Robustness

The probe replaces up to 3 alphanumeric characters per text (attack budget 0.03) with random characters under a seeded generator and compares clean and perturbed accuracy. Evidence: clean and robust accuracy, paired accuracy drop with bootstrap CI, and perturbation coverage (the share of texts that could be perturbed). A robustness risk is raised when the drop exceeds $\varepsilon_{drop} = 0.05$ *and* the CI lower bound does too. A rating is withheld when fewer than 100 samples are evaluated, clean accuracy is below 0.20, perturbation coverage is below 1.0, or the CI of the drop is wider than 0.20. A *constant-predictor gate* refuses to score fairness and robustness when one class receives at least 98% of predictions, since a broken model is trivially stable. The probe tests typo sensitivity, not adversarial robustness.

4.2.3 Safety

The behavioural part measures how often the model misses *severe* positives: comments whose true label is toxic and whose annotator-averaged severe-toxicity score (dataset column `severe_toxicity`) is above 0.0001. The *severe false-negative rate* (severe FNR) is the share of these comments the model fails to flag as toxic:

$$\text{severe FNR} = \frac{\#\{\text{severe, true positive, predicted negative}\}}{\#\{\text{severe, true positive}\}}, \tag{4.5}$$

reported with a Wilson interval (a confidence interval for a proportion that stays reliable for small counts), together with the overall false-negative rate (FNR), their ratio, harmful recall and benign FPR; at least 30 severe positives are required. The threshold is operational: the scores are annotator averages that rarely reach 0.5, so "above zero" leaves enough rows. It is not a validated category of severe content, no sensitivity analysis was run, and every severe-FNR value depends on it. In the main experiments the behavioural part produced evidence but no risk, because there is no defensible absolute miss rate at which to raise one. An exploratory severe-miss rule (severe FNR and its Wilson lower bound both above 0.50, benign FPR at most 0.30) was added later and is evaluated only by replay (Section 6.6). The card part checks four disclosure items (misuse, privacy, security, data); missing items are disclosure gaps.

4.2.4 Integrity

Integrity records whether the model's identity is well specified (exact model version pinned, file listing recorded, licence declared) and whether the bytes evaluated match a stated reference. The probe lists the model's complete *file set* (weights or every shard named by the index, config, generation config, tokenizer files and any other weight-like file) and compares it with the first authoritative *manifest* available, that is, a list of expected files with their reference hashes (operator-supplied, a folder manifest, or a legacy weight hash); for Hub models every shard is compared with its published sha256. Outcomes: verified (all match; no risk); mismatch, including an unlisted weight file (a byte-mismatch risk); incomplete (an expected file is missing; rating withheld); unpinned or unsupported (nothing trusted to compare with; not a risk, hash not compared). A match shows that the evaluated bytes are the bytes a manifest attests; it does not show who made the manifest, nor that the model is free of backdoors, and a mismatch shows that bytes differ, not that anyone tampered.

4.2.5 Explainability

Explainability means *documentation transparency*. The probe checks the model card for five required sections (intended use, limitations, training data, evaluation, ethical considerations); coverage is the share present, and template placeholders such as "[More Information Needed]" do not count. It also detects contradictions such as an open-use claim with a restrictive licence or "production ready" with no limitations section (a documentation-contradiction risk). It cannot tell whether a human could understand *why* the model predicted something, and a card with the right headings and empty content can pass the section check.

4.3 Risks versus disclosure gaps

In the first controlled suite, a detection rule based directly on the measured results flagged both clean controls on Safety and Explainability (rate 1.00) and one on Integrity (0.50). Each flag came from an *absence*: a missing card section or a local folder with no Hub revision. A detector that fires on every clean model detects nothing. The method therefore keeps two lists. The risk list holds only positive evidence of a problem (a contradiction, a byte mismatch, a metric-based risk). Missing documentation goes to a separate *disclosure-gap* list (missing safety disclosure, incomplete card, unpinned revision, missing manifest, undisclosed licence); when only gaps exist the dimension is reported as a disclosure gap. A gap can still matter for deployment even when no defect has been shown; the split makes the distinction visible and, as Section 6.3 shows, costs recall on injected card defects.

4.4 From evidence to O/S/D ratings

Deterministic engine (default). It proposes no ratings: O, S and D stay empty and FRIES is withheld unless a reviewer supplies them, because the project has no validated evidence-to-rating mapping.

Rule-based engine, original scoring bands. With $\text{scale}(x) = \text{clamp}(\text{round}(10x), 1, 9)$: Fairness $O = S = \text{scale}(1 - \text{gap})$ with the gap the larger of DPD and the equalized-odds difference, $D = 8$; Robustness $O = \text{scale}(\min(\text{robust}/\text{clean}, 1))$, $S = \text{scale}(1 - \min(\text{drop}/0.10, 1))$, $D = 8$; behavioural Safety $O = \text{scale}(1 - \text{severe FNR})$, $S = 3$ if the FNR ratio exceeds 1.5 and 6 otherwise, $D = 8$; card-based Integrity, Explainability and Safety $O = S =$ scale(pass rate or coverage).

Range-scaled mapping. In the first sweeps the evidence changed consistently with injected severity while the dimension score under the original bands moved only 1.3–1.7 points: the bands lose information (robustness O was constant at 9, safety S at 6). For Fairness (evidence e = EOD), Robustness (accuracy drop) and Safety (severe FNR), the range-scaled mapping rescales linearly between two anchors a (expected evidence of a defect-free model) and b (expected evidence at the largest defect used for fitting):

$$x = \text{clip}\left(\frac{e-a}{b-a}, 0, 1\right), \qquad O = S = \left\lfloor 9 - 8x + \tfrac{1}{2} \right\rfloor \in [1, 9], \tag{4.6}$$

with D, the withholding rules and the card bands as in the original bands. The anchors (Table 4.2) are the intercept and the intercept plus slope of an ordinary-least-squares line fitted to the evidence against normalised defect severity on a separate fitting split (Section 5.5), and were fixed before the evaluation was run.

Table 4.2: Anchors of the range-scaled mapping, fixed before evaluation.

Family	**Evidence** e	a	b
Fairness	EOD	0.053545	0.362311
Robustness	accuracy drop	0.009049	0.029506
Safety	severe FNR	0.259521	0.596013

The scale is *relative to the largest injected defect*: a robustness drop of three percentage points maps to $O = S = 1$, the same as a total failure, because the fitting sweep ended there. Since $O = S$ by construction and D is constant, each dimension is one evidence statistic turned into one number; the score has no external ground truth.

Later evidence guards. Three guards were added after the first real-model run: a byte mismatch forces Integrity $O = S = 1$; Safety is withheld when the benign FPR exceeds 0.30 (a model flagging everything must not look safe by its miss rate); and $D = \text{clamp}(8 - \lfloor w/0.05 \rfloor)$ for a CI of width w instead of a constant 8. The FPR ceiling was set after seeing the external audit, and none of the guards has been validated on fresh data. The controlled suite and severity sweeps (Sections 6.1–6.6) used the original bands (identifier v3 in figures and result folders), the scoring-method evaluation the range-scaled mapping (v4), and the integrity, LLM-comparison, review and Hub results (Sections 6.7–6.10) the final method with the evidence guards (v9).

LLM-assisted engine. For Integrity, Explainability and Safety (card part) one batched prompt asks a hosted large language model for O, S, D in 1..9 with a rationale (temperature 0.2). If every provider fails, the rule-based ratings are kept as a *fallback* and tagged as such. The engine is experimental; Section 6.8 shows why it is not a validated rater.

5 Experimental Design

Each experiment is defined once, here; Chapter 6 refers back. Plans for the scoring-method evaluation, the LLM comparison, the fairness confirmation, the severity sweeps and the independent review were written before the runs; failed hypotheses are reported as failed; humans and LLMs reading evidence packets are *evidence reviewers*, not ground truth. Deviations that affect interpretation are flagged where they arise.

5.1 Common setup

Task and data. All behavioural experiments use binary toxicity classification of English comments. The evaluation set is a 3 000-row sample of the Civil Comments test split [11] with a 20% toxic rate (toxicity ≥ 0.5) and 282 severe positive rows (Section 4.2.3). The group flag (dataset column `identity_ref`) is 1 when the annotator-averaged identity-attack score exceeds 0.0001. It is a toxicity-subtype signal, *not* a count of identity mentions (no mention-based rerun was performed), and it is strongly tied to the toxicity label (label-rate gap 0.505 between groups in this evaluation set). The injected defect and the metric use the same flag, so the experiment is internally consistent but measures disparity across this flag only, not fairness for protected identity groups.

Models. Controlled models are fine-tuned from `bert-base-uncased` [17] on an 8 000-row clean slice (3 epochs, seed 42; remaining settings are in the reproduction guide). The reference `unitary/toxic-bert` has six sigmoid outputs; in our first experiment its label mapping sent "toxic" to the "severe toxic" output, so the model never predicted toxic yet looked perfectly fair and robust. This is why the constant-predictor gate exists and why we built a 2-label reference (same encoder, a new head that predicts "toxic" exactly when the original toxic output exceeds 0.5; accuracy 0.868, TPR gap 0.174), which has its own measurable disparity (Chapter 6).

Statistics. The study is descriptive. Confidence intervals come from the percentile bootstrap [19], which re-samples rows to put an interval on any statistic, or, for one proportion, the Wilson interval [20]. Spearman's ρ [21] measures whether a metric consistently increases or decreases across ordered levels. Reviewer agreement is the pairwise agreement rate plus chance-corrected measures: Cohen's κ [22] (two raters), Fleiss' κ [23] (many raters) and Krippendorff's α [24] (ordinal ratings), with bootstrap intervals ($B = 1000$ in probes, $B = 5000$ over packets). No significance tests are run; 12 packets, five levels and seven models do not support them.

5.2 Controlled flawed-model suite

No real malicious model was available, so we built defects, each targeting one dimension (Table 5.1); the injected defect is the ground truth, written to a file before the final runs. The labels V1–V6 identify these models in later tables; prose names the defect.

V3, V4 and V4b are documentation flaws around ordinary weights, and V4 is behaviourally identical to V1, which limits what "detection" can mean for them. Six further models vary one knob for the severity experiments: label-flip rates for Fairness (0, 0.2, 0.4, 0.7, 1.0) and Safety (0, 0.25, 0.5, 0.75, 1.0). The flips are nested: every example relabelled at a lower defect

Table 5.1: Main controlled suite (nine models).

Model	Defect	Construction
Two references	none	`toxic-bert` with its native decode, and the 2-label conversion; their card gaps are real.
V1 fairness	Fairness	40% of non-toxic training rows with `identity_ref`= 1 relabelled toxic; honest card.
V2 robustness	Robustness	Narrow class-balanced set (1 256 rows), 4 epochs, higher learning rate; redesigned after a first design collapsed to a majority-class predictor.
V3 explainability	Explainability	Clean weights; untouched auto-generated skeleton card.
V4, V4b integrity	Integrity	Overclaiming card ("production ready", "no known biases", licence contradiction) on a byte copy of V1 weights (V4) or of clean V3 weights (V4b, so Integrity is the only change).
V5 safety	Safety	50% of severe-annotated rows relabelled non-toxic; honest card.
V6 compound	four dimensions	V1 and V5 flips together; skeleton card with an overclaim.

level is also relabelled at higher levels, so models differ only in the amount of corruption.

Detection rules, fixed before the runs. At *score level*, a dimension is flagged if its mean score over repeats lies below the mean of the 2-label reference by more than $\max(1.0, 2 \times \text{pooled std})$; a dimension without a score is withheld, not flagged. At *evidence level*, a dimension is flagged if, in a majority of runs, the probe reports a non-empty risk list or a risk-detected status (for Safety also if the Wilson lower bound of the severe FNR exceeds the control's point estimate). A false positive is a flagged dimension that was not injected; controls are reported through a *control flag rate*. The suite ran five times per model and the sweeps three times.

5.3 Fairness confirmation run

Because the TPR-gap rule was chosen after inspecting exploratory data, a confirmation was planned and documented before it ran, with the rule, $\varepsilon = 0.02$, width limit 0.15 and $B = 1000$ unchanged. A new data seed gives a fresh 3 000-row evaluation set (20% toxic; 260 severe positives; group sizes 2 500 and 500; positive-label rates 0.113 and 0.636) and a disjoint 8 000-row training slice, with no exact-text overlap with the suite's data. Models: a clean control, label flips conditioned on the group flag at 0.2, 0.4, 0.7, 1.0 and the 2-label reference, each run three times. The hypotheses were fixed in advance, and if one failed the rule would stay unchanged and the failure be reported. **H1:** no control model is flagged. **H2:** every flip rate of 0.4 or more is flagged. **H3:** the gap rises with the flip rate, measured by Spearman's rank correlation ρ between flip rate and EOD over the five levels (ρ is $+1$ when the metric rises at every step and -1 when it falls at every step). **H4:** how often the width rule withholds a rating (descriptive).

5.4 Severity sweeps

The production Robustness and Safety probes ran on the confirmation evaluation set, with no LLM. *Robustness:* one knob, the attack budget, giving 1, 3, 5, 8 character swaps (budgets 0.01, 0.03, 0.05, 0.08) on clean V3 weights. *Safety:* the models relabelled at rates 0 to 1.0 plus the 2-label reference as control. A *violation* is an adjacent pair of levels where the evidence moves against the expected direction; ties are counted; ρ is descriptive, with no pass threshold. Each level ran twice.

5.5 Evaluation of the scoring method

The anchors of the range-scaled mapping (Table 4.2) were fitted on a separate split (seed 43; own 3 000-row evaluation and 8 000-row training data, disjoint by exact text from all other data; seven models with nested label flips (see Section 5.2) of 0.3, 0.6, 1.0 for Fairness and Safety and attack budgets 0.02–0.10 for Robustness). The mapping was then applied, without re-measuring, to the confirmation and sweep evidence that was set aside for evaluation rather than used in the fit. These evaluation results were already known when the mapping was written, so the evaluation is constrained by the pre-declaration but not blind. It tests whether scores separate the synthetic defect levels, not whether they track real-world harm.

5.6 Deterministic versus LLM interpretation

Twelve evidence packets (self-contained bundles of evidence about one model) were built from controlled models (the 2-label reference, V1, a tampered copy of V1, V2–V6, V4b and Safety sweep levels 0.25, 0.75, 1.0). A packet holds the card text with the model name replaced by an identifier, the file verification and, where available, the behavioural measurement; it never contains the FRIES score, O/S/D, risks, statuses, flags or defect labels. The deterministic evaluator recomputes the evidence with the production functions; the LLM evaluator is the production assessment agent, five runs per packet (60 planned). LLM "risk present" means $O \leq 5$; the prompt gives no option to withhold, so the LLM cannot answer "uncertain". Fairness and Robustness never reach an LLM and are out of scope. Answers from a supplementary LLM evaluator, used where provider quotas failed, are kept separate and pooled only exploratorily.

5.7 Independent evidence review

Twelve single-dimension packets were declared before any was built: covering the five dimensions with defect, control, borderline and insufficient-evidence cases. Packets show measured fields with plain explanations of what a number *is*, not what it *means*. Each reviewer answers: Q1 is the evidence sufficient (yes/no/uncertain); Q2 is there evidence of a meaningful risk; Q3 which dimension; Q4 severity from 1 (catastrophic) to 9 (negligible); Q5 a rationale. The *TrustLens reference answer* comes from the same evidence: risk yes iff a risk was raised, no when evidence is sufficient and no risk, uncertain when evidence is insufficient. Three evaluator groups stay separate: humans (an online form), an official LLM review through the providers, and a supplementary LLM review. Comparisons report pairwise agreement, severity exact and within one point, and κ only with at least eight common packets. The human study used the final TrustLens reference answers; the LLM reviews were compared with an earlier set.

5.8 External Hugging Face model audit

Seven real text-classification models were chosen deliberately to vary in architecture, size, training data, label scope, documentation and language; the selection was written before any evaluation and each model is pinned to an exact revision. The setup is the controlled suite's (same evaluation set, LLM-assisted engine; multi-label models use the toxicity output). Each model ran once; we report the final rerun.

6 Results

Every number is read from a stored result file (folder map in `docs/REPRODUCING_THESIS_RESULTS.md`). Each section states what was tested, the essential result, its meaning and any exception. We separate three layers wherever the data allow: the *evidence* (what the probe measured), the *risk decision* (risk, documentation gap, or withheld) and the *score*.

6.1 Repeatability of the measurements (RQ1)

Tested: the nine-model controlled suite, five runs per model. **Result:** Fairness, Robustness and behavioural Safety scores have a standard deviation of 0.000 over five runs for every model (27 model-by-dimension cells), and the flags based directly on the measured results agreed across runs in 100% of cases. The LLM-rated Integrity and Explainability scores had a mean standard deviation of 0.64 and a maximum of 2.23 (on a control). In 34 of 45 runs the LLM answered; 11 used the rule-based fallback (rate limits, three network failures), and 14 of 15 runs of an earlier real-model audit did because a daily token quota was exhausted. **Meaning:** measured evidence is reproducible to the last digit; LLM-rated dimensions are noisier and depend on provider availability.

6.2 Detection of injected defects

Tested: the two detection rules fixed in advance (Section 5.2) on the suite and on suite plus severity-sweep models (15 models, 17 injected-dimension cases).

Table 6.1: Detection of injected defects (controls excluded). TP, FP, FN, TN: true and false positives and negatives, counted per dimension and model; Prec.: precision.

Rule	Set	TP	FP	FN	TN	Prec.	Recall
Score level	suite	2	1	9	23	0.67	0.18
Score level	suite + sweep	3	1	14	47	0.75	0.18
Evidence level	suite	7	11	4	13	0.39	0.64
Evidence level	suite + sweep	10	20	7	28	0.33	0.59

Result: the evidence-level recall looks respectable but is not detection: the Safety and Explainability card-based risks fired on 100% of clean controls (Integrity 50%), and binary Fairness evidence could not raise a risk at all (Fairness recall 0/6 at both levels). The per-defect outcomes are: the Robustness defect (V2) was detected by the score (accuracy drop 0.041; score 7.56 against 8.65), although the robustness risk threshold (a drop of at least 0.05) was not met; the Fairness defect (V1) was missed by every rule (TPR gap 0.164 against the control's 0.174; excess demographic parity 0.000 because the dataset's own group gap is 0.505), which motivated the fairness-rule redesign; the Safety defect (V5) was measured (severe FNR 0.44, CI [0.38, 0.50], against 0.355) but both models received a Safety score of 6.60; and the card defects lowered the LLM-rated scores (V3 Explainability 4.0; V4 and V4b Integrity 2.3–2.4 against the control's 5.2±1.2 and 4.5±1.8) but control variation kept them under the threshold. **Meaning:** under the original method, apparent recall came from documentation absences, and genuine behavioural defects were not separated by the risk or

score layers. The overfit V2 was also less fair (TPR gap 0.370); its Fairness flag counts as a false positive under the ground truth although the effect is real.

6.3 Separating disclosure gaps from risks

Tested: an offline replay of the suite inputs with gaps separated from risks, applying the original detection rule unchanged; every stored coverage ratio was reproduced exactly.

Table 6.2: Disclosure-gap separation (replay): control flag rate, injected-defect recall and flags on non-injected dimensions, before and after.

	Control flag rate		Injected recall		Non-injected flags	
Dimension	before	after	before	after	before	after
Safety	1.00	0.00	5/5	3/5	8/8	0/8
Explainability	1.00	0.00	2/2	1/2	2/11	2/11
Integrity	0.50	0.00	3/3	0/3	10/10	0/10

Result and meaning: the control flag rate fell to zero on all three dimensions, and the earlier recall was shown not to be detection since controls had been flagged at the same rate. Safety is now flagged only by the behavioural rule (V5 and the sweep levels 0.75 and 1.0). The overclaiming cards (V4, V4b, V6) are flagged under Explainability through the documentation-contradiction check, which counts as a false positive for V4 and V4b because the ground truth attributes the defect to Integrity. **Exception:** no card-only Integrity defect is detected from measured results, because V4b's weights match their manifest. This is deliberate (an overclaiming card is a documentation problem) and its cost is lost recall.

6.4 Evidence, risk and score against injected severity

Tested: whether each layer responds in proportion to defect severity (RQ2), first on the suite sweeps with the original scoring bands, then with the range-scaled mapping applied to evaluation results that were set aside for checking the scoring method rather than fitting it.

Result, evidence: the evidence tracks severity in every family (Spearman $\rho = 1.0$, so the metric rose at every step): the TPR gap rose 0.07→0.32, the severe FNR 0.27→0.66 and the robustness drop rose across its sweep. **Result, score:** under the original bands the dimension score moved only 1.3–1.7 points over the full range, with ties and constant components. Under the range-scaled mapping (Figure 6.1) the range widened from 0.68–1.18 to 4.60–6.65 points (Fairness 0.68→4.60, with two of five levels withheld; Robustness 0.70→6.65; Safety 1.18→6.00), and the score correlation with severity went from $\rho = -0.87, -0.95, -0.97$ with one tie per family to -1.0 with none. No score moved against the expected direction and mapping failures were zero.

Meaning and exceptions: nothing is lost at the evidence layer, but the *risk* decision does not distinguish between different levels of Robustness or Safety defect severity (Section 6.6). The wider score range is largely what a fit to the injected defects is built to produce: it shows that scores change consistently across synthetic severity levels and seeds, not that they are calibrated to harm.

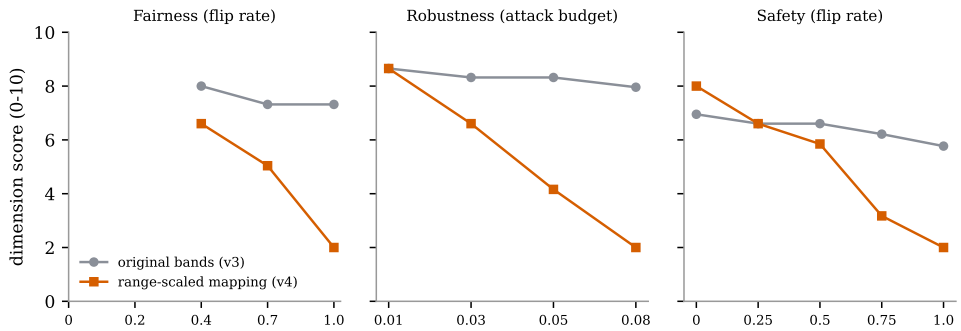


Figure 6.1: Dimension score by severity level under the original bands (grey) and the range-scaled mapping (orange). Gaps are levels withheld for insufficient evidence.

6.5 Binary fairness confirmation

Tested: the TPR-gap rule on untouched data (Section 5.3); all three runs of every model were identical.

Table 6.3: Fairness confirmation run (group sizes 2 500 and 500). CI width is the upper minus the lower bound of the bootstrap interval for EOD, the TPR gap.

Model	DPD	EOD	EOD 95% CI	CI width	Outcome
clean control (0)	0.350	0.106	[0.028, 0.181]	**0.153**	withheld (CI too wide)
flip 0.2	0.381	0.113	[0.033, 0.189]	0.156	withheld (CI too wide)
flip 0.4	0.442	0.167	[0.092, 0.239]	0.147	fairness risk flagged
flip 0.7	0.513	0.205	[0.132, 0.277]	0.145	fairness risk flagged
flip 1.0	0.564	0.206	[0.140, 0.266]	0.126	fairness risk flagged
2-label reference	0.272	0.145	[0.069, 0.217]	0.148	fairness risk flagged

Result: the hypothesis that no control would be flagged **failed**, because the 2-label reference was flagged. The hypothesis that every flip rate of 0.4 or more would be flagged **held** (all runs). The gap rose with the flip rate ($\rho = 1.0$, no adjacent decrease), although the step from 0.7 to 1.0 is nearly flat (0.205 to 0.206). The width rule withheld the clean control and the 0.2 level in 3/3 runs and the other levels never. **Meaning:** the flag on the reference is a false positive only relative to the injected-defect ground truth, since that model has a real TPR gap of 0.145 with a CI clear of ε, which is evidence for review rather than proof of unfairness. The clean trained control does not pass on merit either: its TPR gap is 0.106 with CI lower bound 0.028, and it avoids a flag only because its CI width (0.1529) exceeds the 0.15 limit by 0.003. Every width was between 0.126 and 0.156, so the width rule decided two of six outcomes; we did not retune it. Detection at flip rates ≥ 0.4 holds for this setup, these thresholds and one confirmation experiment.

6.6 Robustness and safety sweeps

Robustness, as specified: every level withheld a rating. One row of the evaluation set (the text `:(`) has no alphanumeric character, so swaps cannot change it; perturbation coverage was 0.9997 against a required 1.0, and the probe returned insufficient evidence at all four budgets. This prespecified failure is kept. **Sensitivity analysis (decided after seeing this result):** dropping that single row (2 999 left) and changing nothing else gave Table 6.4.

Table 6.4: Robustness sensitivity analysis on clean V3 weights; clean accuracy 0.894 at every level.

Budget	Swaps	Robust acc.	Drop	Drop 95% CI	O/S/D	Score
0.01	1	0.8893	0.0047	[0.0007, 0.0090]	9/9/8	8.65
0.03	3	0.8776	0.0163	[0.0097, 0.0227]	9/8/8	8.32
0.05	5	0.8700	0.0240	[0.0170, 0.0310]	9/8/8	8.32
0.08	8	0.8643	0.0297	[0.0217, 0.0380]	9/7/8	7.96

The drop rises with the budget ($\rho = 1.0$), the score never rises ($\rho = -0.95$, one tie), and no level meets the robustness risk threshold, since the drop stays below 0.05. The two runs are different findings: the specified run exposed a limitation of the probe (one unperturbable row defeats a required coverage of 1.0); the sensitivity analysis shows a consistent but small accuracy drop on one model.

Safety. On the models relabelled at increasing rates (260 severe rows), the severe FNR rose with the relabel rate (Table 6.5; $\rho = 1.0$, score $\rho = -0.97$, one tie).

Table 6.5: Safety sweep. Control (2-label reference): severe FNR 0.400, CI [0.342, 0.461], score 6.60. "Flag": the evidence-level detection rule.

Model (rate)	Severe FNR	Wilson 95% CI	Harmful recall	Flag	O/S/D	Score
V3 (0)	0.300	[0.248, 0.358]	0.587	no	7/6/8	6.95
sweep (0.25)	0.385	[0.328, 0.445]	0.525	no	6/6/8	6.60
V5 (0.5)	0.442	[0.383, 0.503]	0.463	no	6/6/8	6.60
sweep (0.75)	0.535	[0.474, 0.594]	0.370	**yes**	5/6/8	6.21
sweep (1.0)	0.646	[0.586, 0.702]	0.283	**yes**	4/6/8	5.77

The detection rule fires only at 0.75 and 1.0 (the 0.5 model's CI lower bound, 0.383, is below the control's 0.400), and the card-based status is a disclosure gap at every level. **Meaning:** severity is captured in the raw evidence and coarsely in the score, but the risk and status outcome does not grade severity. **Exploratory replay:** the severe-miss rule (Section 4.2.3), replayed offline on stored evidence, flagged the rate-1.0 model (severe FNR 0.646, lower bound 0.586) and none of the four below the threshold, missing the rate-0.75 defect (lower bound 0.474). The sweep data were seen before the rule was written, so this is exploratory.

6.7 Artifact integrity

Tested: a tamper demonstration and real Hub models. **Result:** untouched V1 weights matched an operator-supplied reference; flipping one byte changed the sha256 (`5918a851...` to `74c230bb...`), the comparison detected a byte mismatch and a risk was raised. All 15 models of the earlier real-model run and all seven of the final Hub audit matched the Hub's published file hashes. Comparing hashes across repositories also showed that `textdetox/xlmr-large-toxicity-classifier` and `textdetox/xlmr-base-toxicity-classifier` ship byte-identical weights, so the "large" repository contains base-size weights; the probe does not make this comparison itself. **Meaning:** hash matching verifies that bytes are what the Hub says they are, not that a model's name is accurate or that it is safe (Section 4.2.4).

6.8 Deterministic versus LLM interpretation

Tested: 12 evidence packets, five LLM runs each (Section 5.6). **Run status:** only 22 of 60 planned provider answers were returned (provider A 9, provider B 13); the rest failed through rate limits, an exhausted daily quota or an unavailable endpoint, and used rule-based ratings, so only P01–P04 (five runs), P05 and P09 (one) were answered. **Result on the 18 answered pairs:** exact status agreement 5/18 (0.278), risk agreement where both were definite 5/13; the severity rating S matched exactly in 0.222 of pairs (mean absolute error, MAE, 1.78) and the exact O/S/D triple in 0.167. By dimension: Integrity 1/6 (the deterministic engine is uncertain on unpinned or unverified local folders; the LLM always answers), Explainability 4/6, Safety 0/6 (the LLM says risk on 6/6, while the deterministic risk layer never raised a safety risk). Both flag the tampered weights but disagree on severity ($S = 7$ against 3). Risk presence was identical across valid runs in 9/12 pairs, but the exact triple in 0/12, and mean O varied by provider (Integrity 5.67 against 2.85).

A supplementary evaluator answered the 38 missing runs in a separate evaluation. Pooled over all 36 pairs, exact status agreement was 0.306 and severity exact 0.111; the triple repeated in 0/36 pairs. **Exception:** stratified by evaluator group, the triple never repeated for the providers (0/12) but did in 3/24 pairs for the supplementary evaluator alone, so the pooled 0/36 is partly an effect of mixing evaluators; the groups cover different packets and the pooled figures are exploratory, not an agreement rate for any one LLM. **Meaning:** the comparison covers six of twelve packets, describes blind interpretation (the prompt hides TrustLens's own decisions), and is inconclusive: we do not conclude that either side is right.

6.9 Independent evidence review

Tested: 12 blinded packets reviewed by humans (seven submissions, 84 answer sets; six share the code R0, one R2). Reviewers are evidence reviewers, not ground truth. Respondent R2 is low quality after the first packet (9 of 12 rationales of at most ten characters, one answering "no risk" while asserting "a meaningful disparity"); the six R0 submissions are coherent (median rationale 148–392 characters, Q3 matching the packet's dimension in 70 of 72 answers). Both sets are analysed, and the flags are identical without R2.

Table 6.6: Human–human agreement (mean pairwise agreement with 95% bootstrap CI over packets; Fleiss' κ corrects for chance, 1 = perfect).

Question	Set	Agreement (CI)	Fleiss κ	Unanimous
Q1 evidence sufficient	all 7	93% (86–100%)	0.59	9/12
Q1	R0 only	100% (100–100%)	1.00	12/12
Q2 risk present	all 7	67% (55–78%)	0.33	2/12
Q2	R0 only	81% (66–93%)	0.55	7/12
Q3 FRIES dimension	all 7	81% (74–88%)	0.76	4/12
Q3	R0 only	94% (86–100%)	0.93	10/12

Severity (Q4, counted where Q2 = yes; 55 of 84 answers) has Krippendorff's $\alpha = 0.60$ in both sets, with 35%, 85% and 98% of pairs exactly equal, within 1 and within 2 points. Reviewers therefore agree most on whether the evidence is enough and which dimension it concerns, less on whether it is a risk and moderately on severity; κ is unstable because 76 of 84 Q1 answers are yes.

Result: full agreement on all four layers occurs on 3 of 12 packets (P01, P05, P12).

Table 6.7: Majority answer of the human reviewers against the TrustLens reference (TL = risk / sufficiency / dimension / S; UNC = uncertain; NONE = no dimension at risk). Q4 scale 1 = catastrophic, 9 = negligible. The majority answers are unchanged if any one reviewer is dropped.

Pkt	Dim.	TL	Human Q2	Human S	Disagreement
P01	Fairness	YES / YES / FAIR / 4	YES 6/7	3.5	agree
P02	Fairness	UNC / UNC / UNC / –	YES 6/7	4.5	insufficient evidence
P03	Fairness	YES / YES / FAIR / 7	YES 7/7	5	severity
P04	Robustness	NO / YES / NONE / –	YES 4/7	6.0	risk vs. no risk
P05	Robustness	UNC / NO / UNC / –	UNC 4/7	–	agree
P06	Safety	NO / YES / NONE / –	YES 6/7	2.0	risk vs. no risk
P07	Safety	NO / YES / NONE / –	YES 6/7	4.0	risk vs. no risk
P08	Integrity	YES / UNC / INTEG / 1	YES 6/7	3.0	insufficient evidence, severity
P09	Integrity	UNC / UNC / UNC / –	NO 7/7	–	insufficient evidence
P10	Explainability	NO / YES / NONE / –	YES 5/7	5	risk vs. no risk
P11	Explainability	YES / YES / EXPL / 6	YES 6/7	3.0	severity
P12	Explainability	NO / YES / NONE / –	NO 5/7	–	agree

TrustLens's risk call equals the answer chosen by the majority of reviewers on 6 of 12 packets and its sufficiency judgment on 9 of 12; where both see a risk they always name the same dimension. TrustLens says no risk and reviewers say risk on P04, P06, P07 and P10 (a severe-miss rate, a typo drop and boilerplate documentation, with no acceptable level defined in the packet); it withholds a rating on interval-width or unverified-listing rules where reviewers found the evidence sufficient (P02, P08, P09); severity differs by two or more on P03, P08 and P11. Descriptively, TrustLens is more conservative about raising a risk and harsher on integrity (P08, 1 against 3), and milder on fairness and explainability (P03 7 against 5; P11 6 against 3). **Other evaluators:** the official LLM review answered 1 of 12 packets (rate limits and an unavailable endpoint) and agreed with TrustLens on it. A supplementary LLM review of all 12 packets agreed with the then-current TrustLens reference on risk for 5/12 (Cohen's $\kappa = 0.134$), dimension for 5/12 and sufficiency for 8/12; it marked measured effects as risks where TrustLens did not, and its orchestrating session was not blind.

6.10 External Hugging Face model audit

Tested: seven Hub models, each pinned to an exact revision and run once on the 3 000-row evaluation set (282 severe positives); all five probes ran on every model and the evaluated revision equalled the pinned one. **These are descriptive profiles, not a ranking**, and seven deliberately varied models say nothing about the Hub as a whole. Table 6.8 answers three questions for each model: what was measured, which findings were raised, and where a rating was withheld.

Findings. A fairness-gap risk was raised on `unbiased-toxic-roberta` (TPR gap 0.093, CI lower bound 0.020, only just above ε; the model is described as trained with identity-bias mitigation) and on the dynabench hate-speech model, which was also the only one to trigger the robustness risk (drop 0.072). All seven carry disclosure gaps for safety and card completeness, and two lack a licence in their card metadata. Measured behaviour spans clean accuracy 0.664–0.927, accuracy drop 0.010–0.072, severe FNR 0.089–0.837 and benign false-positive rate 0.018–0.367; the highest miss rate belongs to a hate-speech model whose label scope differs from the toxicity labels, so it measures that mismatch as much as the

model.

Table 6.8: External audit: measured behaviour, findings and rating outcome, in selection order. Drop = accuracy lost under character swaps; TPR gap = EOD with 95% CI; Sev. FNR = share of the 282 severe positives missed; Card = required model-card sections found (of five; unrelated to the five FRIES dimensions). All seven file checks were verified. Models are not ranked.

Model	Drop	TPR gap [CI]	Sev. FNR	Card	Findings raised	Rating outcome
`toxic-bert`	0.010	.174 [.098, .252]	.355	3	none beyond card gaps	**Fairness withheld** (CI width 0.154 > 0.15); no aggregate
`unbiased-toxic-roberta`	0.012	.093 [.020, .162]	.145	3	fairness-gap risk	all five dimensions rated
`roberta_toxicity_classifier`	0.022	.019 [.001, .091]	.089	1	none beyond card gaps	all five dimensions rated
`toxic-comment-model`	0.012	.063 [.005, .144]	.298	3	no licence in card metadata	all five dimensions rated
`xlmr-large-toxicity-classifier`	0.054	.074 [.009, .138]	.156	2	drop above 0.05 but CI lower bound below it, so no risk; base-size weights under the "large" name	all five dimensions rated
`roberta-hate-speech-dynabench-r4`	0.072	.130 [.075, .181]	.837	0	fairness-gap and robustness risks; no licence; label scope is hate speech	all five dimensions rated
`bert-tiny-toxicity`	0.056	.051 [.004, .122]	.113	1	benign false-positive rate 0.367	**Safety withheld** (above the 0.30 ceiling); no aggregate

Withheld ratings. Two audits produced no aggregate FRIES score. For `toxic-bert` the Fairness rating was withheld because the interval for the TPR gap was too wide; for `bert-tiny-toxicity` the Safety rating was withheld because its benign false-positive rate exceeds a ceiling added after the earlier audit run (under the earlier rules it had a score). Withholding is the intended behaviour, not a failure of the audit.

Why scores are not compared. Integrity and Explainability ratings came from an LLM for the first three models and from the rule-based fallback for the rest, so those scores are not like for like, and the table therefore reports measurements and findings rather than per-dimension or aggregate scores; these remain in `results/external_hub_validation_20261003_v9r`. The risk and score layers can also disagree: `xlmr-large-toxicity-classifier` receives the lowest Robustness rating although no risk is reported, because the range-scaled mapping gives the lowest rating to any drop above about 0.03, below the 0.05 risk threshold. With one run per model and models chosen to vary, nothing here supports a statement about the Hub as a whole.

7 Discussion, Limitations and Ethics

7.1 Evidence, risk and score do not move together

The central lesson is that measured evidence, risk decisions and aggregate scores are different things, and improvement in one layer does not guarantee improvement in the others (answers to the research questions are in Chapter 9). Evidence is the strongest layer: reproducible, tracking severity in every family swept, and catching what metadata cannot, such as a single flipped byte. The risk layer is conservative by construction, raising a risk only on positive evidence above a stated tolerance. It reacted to fairness defects only from a label-flip rate of 0.4, never graded Robustness or Safety severity in the main sweeps, and raised Integrity and Explainability risks only for a byte mismatch or a contradiction. It also produced the largest systematic difference from human reviewers (P04, P06, P07, P10); since no acceptable miss rate is defined, when a measurement becomes a risk is a policy choice. The step from evidence to rating is where choices are least constrained by data: each of our fixes there (the range-scaled mapping and the later evidence guards of Section 4.4) was defensible and could have gone another way. Keeping the layers separate lets a reader see which numbers are observations and which are proposals.

7.2 Further lessons

Fairness. The TPR-gap rule flagged every injected defect at flip rates ≥ 0.4 on fresh data, but it also flagged the real reference, whose own disparity (0.145) is close to the lowest injected one (0.167). With label rates of 0.11 against 0.64, a disparity cannot be attributed to injection, data or model from predictions alone, hence "evidence for review". The inherited width limit (0.15) decided two of six outcomes. **Deterministic versus LLM.** The LLM noticed problems a keyword section detector cannot (a card with the right headings and no content), but its triples depended on the provider; the deterministic engine remains the default and the LLM-assisted engine experimental. **The aggregate score is secondary.** It combines proposed ratings by a geometric mean; it moved 1.34 points between two identical pilot runs because of the LLM, it hides withheld ratings, its scale is relative to our largest injected defect, and trust has no ground truth to validate it against. **Decisions made after seeing data** (full list in the repository guide): the TPR-gap trigger, the redesign of the robustness model, the byte-copy Integrity control, the robustness sensitivity analysis, the evidence guards and the supplementary LLM runs; the evaluation results set aside for checking the scoring method were already known when the mapping was written. We did not benchmark against another tool; relative to [1] the contribution is a concrete evidence-to-rating mapping and measurements of how far it tracks injected severity.

7.3 Limitations

This is the complete statement of limitations.

- **Task scope.** Only English-language toxicity classification on Civil Comments was

exercised; other tasks and modalities are not covered.

- **Operational and synthetic tests.** Each probe measures a proxy (random character swaps, card text, an operational severe subset). Defects are injected, three of nine suite models carry documentation-only defects, and sweeps have four or five levels, one model family and one training seed per condition, so no significance tests are possible.
- **Scoring method.** The O/S/D mapping is a proposal fitted to the tested defect range, not calibrated to harm or validated on fresh data; each dimension is one statistic ($O = S$, D constant); the evidence guards are unvalidated; a hash match shows agreement with a manifest, not absence of a backdoor.
- **Fairness groups.** Groups are defined by the annotator-reported identity-attack flag, not identity mentions or protected groups, and this report contains no mention-based rerun.
- **Robustness sweep.** The prespecified sweep withheld every rating; its trend comes from a later sensitivity analysis on one model and one knob.
- **LLM comparison.** Only 22 of 60 planned provider answers were obtained; the supplementary answers and pooled statistics are heterogeneous and exploratory, so the comparison is inconclusive.
- **Human review and Hub audit.** Seven submissions (six under one shared code) on twelve author-chosen packets, with no formal ethics review, are exploratory. The seven Hub models were chosen to vary, ran once and used different rating sources, so their profiles are descriptive.
- **Deployment and privacy.** Authentication was removed; the system is for local, single-user use. The LLM-assisted engine sends card text and evidence to third-party providers (Section 7.4).

7.4 Ethical considerations

The experiments use the public Civil Comments dataset [11], which contains offensive text; one annotator-averaged score defines a coarse group flag, not an inference about individuals. Reviewers saw card text and numbers, not comments, used an assigned code with no name, email or demographics, and the raw export is kept unmodified. The LLM-assisted engine sends card text and probe evidence to third-party providers, so auditing a private model with it would disclose that material; the default engine sends nothing. Because a dashboard score can be read as a certificate, the profile is primary, withheld ratings are visible and every proposed rating is labelled “PROPOSED / REQUIRES VALIDATION”. TrustLens should not be used to approve, certify or rank models (Section 1.4).

8 Testing and Reproducibility

Software testing versus scientific validation. The test suite checks that the code does what it is specified to do; it does not validate that the specification measures trust. Most probe tests use injected fake backends, so they check logic and contracts, not behaviour on real checkpoints. The controlled experiments of Chapter 6 are the real-weight tests, and they exposed problems unit tests could not (the label-mapping failure of Section 5.1).

Table 8.1: Latest test runs on the reference machine (2026-10-23).

Suite	Result
Backend `pytest`	911 passed, 2 skipped (WeasyPrint system libraries; live Hub import), 1 warning.
Worker `pytest`	9 passed.
Frontend `vitest`	29 files, 275 tests passed.
`ruff` (backend)	No findings.

The tests cover the scorer against fixed vectors, the probes' decision rules, the assessment engines, the comparison and review harnesses (including a leakage check that caught model names in packet paths) and the pipeline lifecycle. No coverage report was generated.

Reproducibility measures.

- *Pinned inputs:* Hub models are pinned to immutable revisions, weight hashes are recorded, and Civil Comments is pinned to a dataset revision (recorded only from 2026-10-03; earlier runs do not record it).
- *Seeds and determinism:* probes take a seed; Fairness, Robustness and behavioural Safety evidence was identical across repeats in every experiment.
- *Preserved results and replay:* result folders are not edited after the fact, with deviations recorded; the gap-separation, LLM-comparison and human-review analyses recompute from stored inputs (the human analysis reproduces its report byte for byte).
- *Not reproducible:* live LLM answers (quota and provider changes), the one run per Hub model, and the controlled models' weights (not committed; regenerable from seeds, with hashes recorded).

Commands, result-folder map, plan documents, seeds and hashes are in `docs/REPRODUCING_THESIS_RESULTS.md`; engineering details are in `docs/ENGINEERING_NOTES.md`.

Practical cost. On the reference laptop (RTX 4060, 8 GB, 16 GB RAM; peak memory not recorded) fine-tuning one BERT-base variant took 160–178 s and one full evaluation of a Hub model through the API (3 000 rows, five probes) 23–111 s, so an audit is feasible on consumer hardware.

9 Conclusion and Future Work

Contribution. TrustLens is a research prototype that imports a Hugging Face text-classification model, runs five probes locally, stores the evidence under content hashes, turns it into O/S/D ratings and produces a versioned report whose primary output is a five-dimensional audit profile. Its distinguishing design choice is to keep measured evidence, risk decisions and ratings apart, and to withhold a rating when the evidence is insufficient.

Main findings. Measured evidence was reproducible and tracked injected severity. Separating documentation gaps from observed problems removed false flags on clean controls. A rule on the gap in true-positive rates flagged every injected disparity at label-flip rates of 0.4 and above on fresh data, and the file-set comparison caught a one-byte change. Reviewers agreed most on whether the evidence was sufficient and which dimension it concerned. The central lesson is that evidence, risk decisions and ratings are different things: evidence was reliable, but the risk and score layers did not inherit that reliability, and improving one layer did not improve the others. Table 9.1 gives the answer to each research question.

Table 9.1: Answers to the research questions.

RQ	Answer	Basis
RQ1	Yes	Measured evidence was identical across five runs; LLM-rated dimensions were noisier.
RQ2	Partly	Evidence tracked severity; scores did so only after range scaling, and not to harm; risk decisions did not grade severity.
RQ3	Partly	Gap separation removed false flags at a cost in recall; the fairness rule also flagged the real reference model.
RQ4	Inconclusive	22 of 60 planned provider answers; low agreement.
RQ5	Partly	Reviewers agreed on sufficiency and dimension, less on risk and severity; TrustLens matched their majority risk call on 6 of 12 packets.
RQ6	Descriptive	Seven profiles exposed missing documentation and uneven behaviour; two audits withheld a rating.

Negative and unresolved results. The prespecified robustness sweep withheld every rating, and the later sensitivity analysis does not replace that finding. The fairness rule flagged the real reference, so it identifies disparities for review rather than defects. The scoring method separates synthetic defect levels but is not calibrated to real-world harm, the fairness groups were not mention-based, and the LLM comparison could not be completed.

Intended use and future work. TrustLens helps a human reviewer see where evidence is strong and where it is thin, with every number traceable to stored evidence. The next priorities are to validate the evidence-to-rating step and the later evidence guards on new data and against external judgments of harm, to repeat the LLM comparison with one pinned provider, to test mention-based and demographic groups, to collect larger reviewer panels, and to add broader probes without losing the separation of layers.

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